LEBANESE UNIVERSITY

Faculty of Engineering - Branch III • Department of Mechanical Engineering • Academic Year 2025/2026

Individual Contribution Summary & Personal Reflection

Student Name:	Yehia Moslem (ID: 6824)	**Course:**	Sensors
Project Title:	Smart Solar Tracking & Self-Cleaning System	**Instructor:**	Dr. Mohamad Aoude
Group:	Group C4	**Role focus:**	Lead Hardware, Sensor Integration & Calibration

STATEMENT REGARDING GIT COMMIT HISTORY

Please note that because our team completed this prototype entirely on-site in the university, all physical assembly, wiring, circuit debugging, and iterative code development were conducted collaboratively on a single local workstation. Consequently, a traditional Git commit history is unavailable; however, this document serves as a comprehensive individual assessment detailing my primary hands-on contributions, hardware implementation, and verification metrics.

1. INDIVIDUAL CONTRIBUTIONS & HARDWARE LEADERSHIP

As the lead hardware engineer for Group C4, I assumed full responsibility for translating our project's system architecture into a functional physical prototype. This involved the complete hardware execution path: designing the physical component layouts, executing the breadboard wiring, establishing power isolation circuits, and systematically integrating, calibrating, and troubleshooting every sensor and actuator. Specifically, I led the physical setup of the **dual-axis solar tracking platform** and the **automatic fluid-cleaning assembly**, ensuring that all subsystems could safely interface with our central Arduino Uno microcontroller.

- **Dual-Axis Tracking Array:** I wired and physically aligned the four Light Dependent Resistors (LDRs) in a voltage-divider topology with $10\ k\Omega$ resistors, designing and constructing the physical opaque barriers between them. I mounted and centered the horizontal (Azimuth) and vertical (Elevation) SG90 servo motors on the 3D-printed conical base and frame to facilitate precise multi-axis positioning.
- **Self-Cleaning & Fluid-Safety Subsystems:** I integrated the TCRT5000 infrared reflective sensor above the panel surface to serve as our optical dust-detection mechanism. To handle the automatic cleaning cycle, I wired a 5V submersible pump via a 2-channel electromagnetic relay module to isolate the inductive pump load from the Arduino, sourcing power from an external battery pack to prevent voltage sags on the MCU.
- **Safety & Feedback Instrumentation:** I structured the water tank on a 3D-printed load cell frame, connecting a 5kg load cell to a 24-bit HX711 analog-to-digital amplifier. Furthermore, I integrated a YF-S201 Hall-effect flow sensor downstream of the pump to verify actual water movement and added a DHT22 digital sensor to monitor ambient temperature and relative humidity.

2. TECHNICAL INSIGHTS: WIRING, MEASUREMENT & SENSOR CALIBRATION

A major part of my focus was mastering how each sensor operates, establishing reliable electrical connections, and developing calibration routines to translate raw electrical values into meaningful physical units:

- **LDR Balancing:** LDRs exhibit subtle natural resistance variations. To balance the tracking array, I measured each LDR's output under a uniform light source and calculated individual normalization scaling factors (k) where $S_{corrected} = k \times S_{raw}$. By evaluating the analog readings from 0 to 1023 (using a $5V$ reference), I implemented a soft-coded threshold of 50 ADC counts to effectively eliminate high-frequency servo jitter and structural oscillations under minor ambient lighting fluctuations.
- **Load Cell & HX711 Calibration:** To monitor water volume, I conducted a multi-point linear regression calibration of the load cell using four reference masses measured on an external precision balance (ranging from $62g$ to $354g$). The resulting calibration equation, $m = a \times x + b$ (where x is the raw 24-bit digital reading, a is the scale slope, and b is the

offset), achieved a root mean square error (RMSE) of only $0.237g$. This enabled us to reliably calculate water volume, since $V_{water}\ (L) \approx m_{water}\ (kg)$, enabling a strict interlocking limit that blocks pump operation below $0.5\ L$.

- **TCRT5000 IR Dust Profiling:** Because the analog IR reflection is highly sensitive to the sensor's physical height, panel color, and ambient lighting, I devised an empirical calibration protocol. We recorded baseline reflections for a clean panel ($S_{clean} \approx 754.3$), light dust, medium dust ($S_{medium} \approx 594.3$), and heavy dust ($S_{heavy} \approx 437.7$). This allowed us to formulate a robust Dust Index: $Dust\ Index\ (\%) = |(S_{current} - S_{clean}) / S_{clean}| \times 100$. By implementing a 20% threshold, the cleaning cycle activates specifically at medium-to-heavy dust loads.
- **YF-S201 Flow and DHT22 Environment Safety:** I mapped the YF-S201 flow sensor pulse frequency to flow rate using the manufacturer's model, $f = 7.5 \times Q\ (L/min)$. If the pump runs and flow $Q < 1\ L/min$ for 3 seconds, the system triggers a "Dry Run" shutdown. Similarly, the DHT22 was calibrated to disable water spraying if temperature $T \geq 35°C$ or relative humidity $RH \geq 70\%$, protecting the panel from thermal shock and preventing water waste due to unreliable dust-sensor readings.

3. PERSONAL REFLECTION & ENGINEERING LESSONS LEARNED

This project was an invaluable hands-on journey that bridged the gap between theoretical sensor physics and practical mechatronic design. Working extensively on the physical implementation, I learned that **sensor integration is rarely plug-and-play**. Every sensor has physical and electrical limitations, such as signal noise, cross-sensitivity, and environmental dependence. For example, resolving the electrical noise generated by the servo motors—which initially caused erratic analog readings across our LDRs—taught me the critical importance of proper circuit isolation, decoupled power lines, and robust physical grounding.

Furthermore, developing the calibration routines for the load cell and the TCRT5000 dust sensor taught me how to handle real-world experimental errors. I learned to apply statistical tools like Standard Deviation and Root Mean Square Error (RMSE) to quantify the reliability of our data. Working as a team on-site also emphasized the value of collaborative engineering: while my partners focused on software design and structural modeling, my role as the hardware lead required me to communicate clear electrical requirements and pin assignments, ensuring our physical wiring matched the logical control flow. Ultimately, this experience has greatly built my confidence in designing, calibrating, and optimizing multi-sensor mechatronic systems.